

Test Coverage Report

Red Thread Game — JaCoCo Coverage Report (Extra)

What is JaCoCo?

JaCoCo (Java Code Coverage) is a tool that measures how much of the codebase is covered by unit tests. Instead of simply saying we have 10 tests, JaCoCo provides an exact percentage per class — for example, 37% coverage for WitnessService and 32% for CaseService. This gives concrete, measurable proof that the tests are exercising real production code.

Why We Used It

Rather than relying on test count alone, JaCoCo provides an objective metric that demonstrates test quality. It generates a visual HTML report breaking down coverage per package, class, method, and branch — making it easy to identify what is tested and what needs more coverage.

Project-Level Coverage Overview

Overall coverage across the full project (46 classes):

 Red-Thread-Game

 Sessions

Red-Thread-Game

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
com.example.redthreadgame.Service		5%		4%	386	400	1,125	1,188	168	179	16	20
com.example.redthreadgame.Controller		0%	n/a	n/a	91	91	130	130	91	91	15	15
com.example.redthreadgame.Advice		1%	n/a	n/a	22	23	22	23	22	23	0	1
com.example.redthreadgame		37%	n/a	n/a	1	2	2	3	1	2	0	1
com.example.redthreadgame.Enums		100%	n/a	n/a	0	7	0	29	0	7	0	7
com.example.redthreadgame.Config		100%	n/a	n/a	0	2	0	2	0	2	0	1
com.example.redthreadgame.Api		100%	n/a	n/a	0	1	0	2	0	1	0	1
Total	6,587 of 7,048	6%	420 of 441	4%	500	526	1,279	1,377	282	305	31	46

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The overall project coverage is 6%. This is expected — the tests cover only a small portion of the project, focusing specifically on CaseService and WitnessService .

Service Layer Coverage Breakdown

Per-service coverage details:

Red-Thread-Game > com.example.redthreadgame.Service

[Source Files](#) [Sessions](#)

com.example.redthreadgame.Service

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
GameSessionService		0%		0%	54	54	157	157	23	23	1	1
OpenAiService		0%		0%	21	21	111	111	7	7	1	1
QuestionService		0%		0%	31	31	121	121	15	15	1	1
SolutionProposalService		0%		0%	33	33	114	114	15	15	1	1
HintService		0%		0%	25	25	66	66	15	15	1	1
SuspectService		0%		0%	23	23	65	65	8	8	1	1
NoteService		0%		0%	25	25	53	53	15	15	1	1
PlayerService		0%		0%	15	15	53	53	9	9	1	1
ProposalVoteService		0%		0%	22	22	54	54	10	10	1	1
CaseService		32%		18%	29	35	60	90	5	10	0	1
WitnessService		37%		35%	21	27	49	80	6	10	0	1
EvidenceService		0%		0%	15	15	42	42	7	7	1	1
ElevenLabsService		0%		0%	22	22	43	43	5	5	1	1
InvitationService		0%		0%	18	18	49	49	8	8	1	1
AdminService		0%		0%	12	12	30	30	6	6	1	1
SessionPlayerService		0%		0%	9	9	21	21	5	5	1	1
CaseSolutionService		0%		0%	7	7	18	18	5	5	1	1
EmailService		14%		n/a	1	2	10	11	1	2	0	1
GameSessionService.new TimerTask() { ... }		0%		n/a	2	2	6	6	2	2	1	1
WhatsAppService		12%		n/a	1	2	4	5	1	2	0	1
Total	5,465 of 5,756	5%	420 of 441	4%	386	400	1,125	1,188	168	179	16	20

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Coverage Summary

Class	Instructions Coverage	Branch Coverage
CaseService	32%	18%
WitnessService	37%	35%